

Nitish Ravisankar Raveendran

+1 240 825 6896 | rmitish@gmail.com | [LinkedIn](#) | [GitHub](#) | rmitish.com

Summary

Robotics Autonomy Engineer skilled in developing end-to-end ROS2 systems for real-world robots. Experienced in perception, planning, and control pipelines using RGB-D and stereo sensors, and in deploying motion planning for constrained tasks. Focused on building robust, production-ready cleaning and service robots through iterative testing and rapid debugging.

Education

University of Maryland, College Park

Aug 2023 - May 2025

M.Eng, Robotics

- **GPA:** 3.79/4.0
- **Coursework:** Machine Learning, AI in Robotics, Sensor Integration, Computer Vision, Sensor Fusion, SLAM

Anna University, PEC, Chennai

Aug 2018 - May 2022

B.E., Electronics and Communications

- **GPA:** 8.94/10

Experience

Anthropilot

Jul 2025 - Present

Robotics Engineer

New York

- Developed ROS2 perception and planning pipelines for autonomous navigation on embedded compute, architecting modular nodes for mapping, obstacle avoidance, and path generation.
- Integrated stereo and LiDAR sensing to enable reliable operation in visually challenging, cluttered spaces improving environment understanding and real-time decision-making.

Cognizant

Feb 2022 - Aug 2023

Software Engineer

Chennai

- Developed and deployed full-stack features using React and Node.js, resulting in a 15% improvement in system responsiveness and showcasing expertise in software engineering

Projects

Bipedal Locomotion Using Reinforcement Learning | <https://github.com/Nitish05/walker>

- Analyzed and optimized model training by implementing and comparing TRPO, PPO, and TQC algorithms to generate adaptive bipedal gaits. Reduced training convergence time by 35% and improved sample efficiency, demonstrating a strong grasp of how architectural choices and optimization techniques impact training performance in complex sequential decision-making problems.
- Tools Used: Python, PyTorch, OpenAI Gym, MuJoCo, ROS 2

Reinforcement Learning for Pursuit-Evasion | https://github.com/Nitish05/ENPM690_final

- Developed a multi-agent DQN framework in ROS2 & Gazebo to investigate emergent behaviors and convergence challenges in a distributed agent system. This work provides insights into training stability and parameter tuning for complex and interactive artificial intelligence models analogous to real-world driving scenarios.
- Tools Used: Python, PyTorch, ROS 2, Gazebo, Docker

StereoSight: Local Planner for Drones | <https://github.com/Nitish05/StereoSight>

- Engineered a real-time perception and planning pipeline in ROS 2, processing stereo camera depth maps into occupancy grids for autonomous navigation. Optimized Dijkstra algorithm for efficient waypoint computation, crucial for low-latency decision making in production systems.
- Tools Used: C++, Python, ROS 2, OpenCV, PCL

TurtleBot3 Vision-Based Path Following | https://github.com/Nitish05/ENPM673_turtlebot_perception_challenge.git

- Designed a robust vision-based perception and control pipeline in ROS2 and OpenCV to enhance autonomous navigation. This system forms a foundational block for end-to-end models, successfully navigating complex paths with varied conditions, increasing route success rates by 40% while maintaining 30 FPS processing.
- Tools Used: C++, Python, OpenCV, ROS2, Linux

Arduino-Based Aimbot with YOLO Object Detection | <https://github.com/Nitish05/AimBot.git>

- Optimized a real-time object detection and actuation system by integrating a YOLOv10 model with an Arduino controller. Reduced end-to-end inference and actuation latency by 50% through efficient model implementation and streamlined data pipelines.
- Tools Used: C++, Python, OpenCV, YOLO, Linux, Arduino

Skills

- **Programming Languages:** Python, C++, C#, JavaScript, SQL
- **Technologies & Tools:** ROS 2, Gazebo, Docker, PyTorch, OpenAI Gym, MuJoCo, OpenCV, PCL, Git, Linux, Isaac Sim, React, NodeJS